Kewei XU

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Hi there !

I am currently a Postdoctoral Researcher in the MANAO team at Inria Bordeaux. My research focuses on material appearance, physically based rendering and 3D reconstruction.

Education

University of Poitiers - XLIM UMR CNRS 7252, Poitiers, France Mar 2023 – Jun 2026

- Ph.D. in Computer Science (Computer Graphics)
- **Advisors:** Mickaël Ribardière, Benjamin Bringier, Daniel Meneveaux

Sorbonne University, Paris, France Sept 2019 – Dec 2022

- Master of Science in Computer Science and Technology (Computer Vision) - *with distinction (mention bien)*

Sorbonne University, Paris, France Sept 2018 – Sept 2019

- Bachelor of Science in Computer Science and Technology - *with distinction (mention bien)*

IUT of Orsay, Paris-Sud University, Orsay, France Sept 2016 – Sept 2018

- University Diploma of Technology (DUT) in Computer Science - *with high distinction (mention très bien)*

Experience

Research Intern, LINEACT CESI – Rouen, France Feb 2022 – Sept 2022

- **Advisors:** Nicolas Ragot, Yohan Dupuis
- Bring computer vision solutions into industrial digital twins by selecting high-score viewpoints to boost recognition reliability and efficiency. We propose a scoring mechanism that chooses optimal views for object recognition and extends to industrial assembly-step recognition; it outperforms random views on small datasets and stays robust under simulated robotic-arm viewpoint offsets up to $10°$. Traditional clustering yields F1 score 0.6, while MobileNet with transfer learning reaches 0.9; for datasets with highly similar classes, image similarity can be fused into the score to improve discrimination.

Software Engineer Intern, Sichuan Normal University – Sichuan, China Apr 2018 – Jul 2018

- Built full-stack features for a student registration system, Implemented form validation, authorization, improving submission reliability and data integrity. Optimized API and database design to reduce latency on frequent operations.

Publications

A Discrete Polydisperse Anisotropic BSDF Model Based on the Micrograin Framework
Kewei Xu, Simon Lucas, Mickaël Ribardière, Benjamin Bringier, Pascal Barla

- Published in *Computer Graphics Forum, Proceedings of Eurographics 2026*

Real Surface Measurement and Virtual Gonioradiometer for Road Appearance Prediction
Kewei Xu, Mickaël Ribardière, Benjamin Bringier, Daniel Meneveaux

- Published in *MAM - MANER London, 2024*

Virtually Measuring Layered Material Appearance
Kewei Xu, Arthur Cavalier, Benjamin Bringier, Mickaël Ribardière, Daniel Meneveaux

- Published in *Journal of the Optical Society of America A, 2024*

View Selection for Industrial Object Recognition
Kewei Xu, Nicolas Ragot, Yohan Dupuis

- Published in *IECON 2022 - 48th Annual Conference of the IEEE Industrial Electronics Society, 2022*

Personal Mini-Projects

A real-time rasterizer little demo based on bgfx Project page

- Implementation of a little real-time rasterization render based on bgfx. Supports orbit/pan/zoom camera via mouse and keyboard, textured object rendering with skybox/environment maps, a selectable lighting pipeline (Blinn–Phong or PBR with IBL), and real-time shadows using shadow mapping.

Implementation of variance soft shadow mapping Project page

- Implementation of Variance Soft Shadow Mapping (VSSM) in OpenGL and compares it with some other shadow rendering techniques like percentage-closer filtering (PCF) and percentage-closer soft shadows (PCSS).

Super-resolution for downscaling on oceanographic fields Project page

- I frame the downscaling of oceanographic fields as a super-resolution task and address it with a deep learning approach. Building on the classical SRCNN. I optimize both the network architecture and the data pipeline, and achieve satisfactory results on the NATL60 dataset.

Skills

Programming Languages: C/C++, Python, GLSL

Softwares: Mitsuba, OpenGL, Pytorch, LaTex, Blender

Languages: English (Fluent), French (Commonly used in daily life), Chinese (Native speaker)

Hobbies: Motion design and Video editing using Adobe After Effects and Photoshop YouTube channel